Probabilistic Solar Irradiance Forecasting for Head Hall, UNB Fredericton.

Mohammed Rahamath Ali[1], Oghenekaro Kennedy O. [1], Zineddine Belmokhtar[1], Julian Cardenas[1]
[1]Electrical and Computer Engineering, University of New Brunswick

ABSTRACT
The growing penetration of renewable energy sources, particularly solar photovoltaics, is increasing the need for accurate, well-calibrated probabilistic forecasts to support unit commitment, reserve scheduling, and real-time operations. This work develops and evaluates a probabilistic day-ahead solar irradiance forecasting framework for Fredericton, Canada, using NASA POWER reanalysis data from January 2022 to October 2025. After data cleaning and hourly retiming, a Seasonal ARIMA (SARIMA) model with 24-hour seasonality is adopted as the statistical baseline, justified by a strong daily cycle and an augmented Dickey–Fuller p-value of 0.001 indicating stationarity after seasonal structure is captured. An advanced Quantile Regression Forest (QRF) model is then trained on a compact 8-feature set (hour_sin, hour_cos, doy_sin, doy_cos, clrsky, kt, irr_lag24, irr_roll24) derived using a correlation matrix and PCA-like eigen decomposition, where the first two components explain 52.6% of the variance. On a representative sunny test day (29-Jul-2025, 24-hour horizon), the SARIMA baseline yields RMSE = 81.1 W/m², MAE = 47.8 W/m², CRPS = 54.20, 90% prediction interval coverage of 100%, and a relatively wide normalized interval (PINAW = 0.669). The QRF model substantially improves point accuracy and sharpness, achieving RMSE = 41.3 W/m², MAE = 21.3 W/m², CRPS = 15.01, and PINAW = 0.154 while maintaining high empirical coverage (95.8%). These results indicate that QRF provides more informative probabilistic forecasts, enabling tighter yet reliable uncertainty bands that can translate into more efficient reserve allocation and operational planning.

1. INTRODUCTION:

The global change toward low-carbon energy systems has elevated renewable energy technologies, particularly solar photovoltaic (PV) power, to a position of critical importance. As nations and institutions make efforts towards reducing greenhouse gas emissions, improving grid resilience, and stabilizing long term operation costs, it is evident that distributed PV installations have emerged as a practical and increasingly accessible solution. Universities, research centers, and commercial facilities have begun integrating rooftop solar systems not only as sustainability initiatives but also as operational assets capable of offsetting significant electricity consumption.

The University of New Brunswick (UNB) is not left out as it installs rooftop PV modules on Head Hall, one of its core academic buildings. These systems provide a measurable contribution toward reducing grid dependence; however, their effectiveness is constrained by the inherent variability of solar irradiance. Solar generation fluctuates due to cloud cover, atmospheric turbidity, seasonal cycle, and diurnal patterns, all of which introduce uncertainty into building-level energy planning.

According to recent literature [1], variability in solar production poses challenges for operational decision making, especially for facilities aiming to maximize self-consumption or control electricity costs. Without expectation of changes in PV output, buildings are forced to procure energy reactively, often drawing power from the public grid at less favorable times. This undermines economic performance, reduces load-matching efficiency, and may lead to underutilization of locally generated clean energy.

Given these challenges, accurate short-term solar forecasting, particularly 24-hour-ahead hourly predictions, is increasingly recognized as a foundational requirement for modern energy management systems. Probabilistic forecasts, which quantify uncertainty rather than providing a single expected value, are especially valuable because they support risk-aware decision-making. For instance, building operators can determine whether additional reserves, battery planning, or adjustments to energy-intensive operations are required based on predicted distributions (P10/P50/P90) rather than a single deterministic curve.

1.1. Problem Statement

Head Hall's solar output is currently monitored, but no forecasting is done, which prevents more responsive energy-use planning with respect to the grid. To address this, we will assess the performance of two models: a Seasonal Autoregressive Integrated Moving Average (SARIMA) model for baseline prediction, and a Quantile Regression Forest (QRF) model as an advanced approach. In a risk or economic decision, it is more important to know the distribution of possible results rather than a single likely one.

1.2. Objective:

This Project aims to develop a 24-hour next-day solar energy forecasting model for the Head Hall rooftop PV system at UNB Fredericton, using past meteorological data on irradiance and temperature. This will be done by:

a) Constructing and comparing SARIMA and QRF models for day-ahead hourly irradiance forecasting,
b) Generating calibrated P50, median, and P90 prediction intervals,

c) Assess predictive accuracy using metrics such as RMSE, MAE, CRPS, coverage probability, and interval width,
d) Demonstrate how probabilistic forecasting improves decision making for solar energy integration and building level energy management.

The scope of this study is limited to one site (Head Hall), one forecasting horizon (24 hours ahead), and four years of hourly meteorological data (2022–2025) sourced from the NASA POWER dataset for Fredericton, NB. This dataset was selected due to its public accessibility, high temporal resolution, and compatibility with rooftop PV simulation workflows. Features include Downward Direct Normal irradiance (DNI), temperature at 2m above ground, specific humidity, and sky clearness index, all of which are known to influence PV output and are commonly used in solar forecasting literature.

2. LITERATURE REVIEW

2.1. Renewable Energy Forecasting in Power Systems

The integration of renewable energy sources (RES) into modern grids has been widely studied due to their variability and stochastic nature. Traditional fossil-fuel generation offers dispatchable stability, whereas solar and wind introduce uncertainty that complicates grid balancing. Accurate forecasting is therefore essential for unit commitment, economic dispatch, and long-term planning. [2] Demonstrated the utility of ARIMA models in predicting renewable shares in Germany, emphasizing the importance of incorporating exogenous variables to improve accuracy. Similarly, [3] applied ARIMA to wind power forecasting in ERCOT, showing tangible financial and operational benefits from improved prediction accuracy. These works underscore the role of statistical time-series models in mitigating uncertainty and enhancing grid resilience.

2.2. Solar Energy Forecasting Challenges

Solar photovoltaic (PV) forecasting presents unique challenges compared to other RES because it depends on meteorological conditions. Cloud cover, atmospheric turbidity, and seasonal cycles introduce rapid fluctuations in irradiance, making deterministic forecasts insufficient for operational decision-making. [4] compared ARIMA with machine learning techniques for solar forecasting, highlighting that while ARIMA captures temporal dependencies effectively, machine learning approaches can better model nonlinearities. However, both approaches struggle under extreme weather variability, suggesting the need for hybrid or probabilistic methods. Furthermore, [5] emphasized that without reliable forecasts, facilities risk reactive procurement from the grid, undermining economic performance and reducing self-consumption efficiency.

2.3. Probabilistic Forecasting Approaches

Recent literature has shifted toward probabilistic forecasting, which provides distributions of possible outcomes rather than single-point estimates. This paradigm supports risk-aware decision-making by quantifying uncertainty. Quantile Regression Forests (QRF), for instance, generate calibrated prediction intervals (P10/P50/P90), enabling operators to plan reserves, battery dispatch, or load adjustments with confidence. Such approaches align with the operational needs of building-level energy management, where uncertainty directly impacts cost optimization and sustainability goals.

2.4. Methodological Advances: SARIMA and Hybrid Models

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model extends ARIMA by incorporating seasonality, making it well-suited for solar irradiance data that exhibit daily and annual cycles. [6] established the methodological foundation for SARIMA, which remains a baseline in time-series forecasting, while [7] refined these techniques, emphasizing the importance of stationarity testing and residual diagnostics. While SARIMA provides interpretable and statistically grounded forecasts, its deterministic nature limits its applicability in risk-sensitive contexts. Hybrid approaches that combine SARIMA with machine learning or probabilistic methods, such as QRF, have been proposed to balance interpretability with uncertainty quantification.

2.5. Building-Level Solar Forecasting

Most forecasting studies focus on utility-scale integration; however, distributed PV systems at universities and commercial facilities are increasingly recognized as critical testbeds. Also, [8] highlighted the growing complexity of electrical networks and the need for robust forecasting at multiple scales. For institutions like the University of New Brunswick, rooftop PV systems serve dual roles: sustainability initiatives and operational assets. This project addresses a gap in building-level forecasting literature, demonstrating how probabilistic methods can enhance local energy management and contribute to broader grid stability.

3. Methodology for Comparative Hourly Solar Irradiance Forecasting

The dataset used in this study was obtained from the NASA Prediction of Worldwide Energy Resources (POWER) project [9], which provides satellite-derived surface meteorology and solar energy variables at an hourly resolution. A request was made for a single point at Fredericton, New Brunswick, Canada (latitude 45.9494°N,

longitude 66.6422°W). The methodology leverages a dual modelling strategy, comparing the performance of the traditional time-series approach (SARIMA) with that of an advanced nonparametric machine learning technique (Quantile Regression Forest, QRF) for day-ahead prediction.

3.1. Data Cleansing and Standardization Pipeline

The raw hourly data set required extensive cleaning to transform raw sentinel values and ensure temporal continuity, hence laying a clean foundation on which later models could be applied. A datetime index was created, and the resulting series spans from 2022-01-01 00:00 to 2025-10-24 23:00 continuously. To obtain a strictly regular hourly grid, the timetable was re-timed to an hourly frequency by retime. This step ensures that later time-series models-SARIMA-see a regularly spaced series even if a few hours were missing in the original file.

After the dataset was re-timed to a continuous hourly index, missing values were handled in two steps: first, linear interpolation was used to smoothly bridge small gaps, and then nearest-neighbor imputation was used to resolve other isolated NANs. Ensuring a complete training series in this way helps to keep the physical validity of solar irradiance values non-negative; any negative entries coming either from interpolation or raw data were clipped to. This preprocessing step ensures that forecasting models run on consistent, realistic inputs and avoid learning physically impossible negative solar power.

3.2. Exploratory Data Analysis (EDA) and Feature Engineering

The raw hourly DNI irradiance series for Fredericton from January 2022 to October 2025 shows a strongly periodic but highly variable structure, shown in Figure 1. On the yearly scale, the series follows the solar cycle, with higher peaks in summer and almost no irradiance in mid-winter. Superimposed on this smooth seasonal pattern are large high-frequency fluctuations caused by cloud cover and short-lived weather systems, which motivates a forecasting framework that can handle both strong seasonality and substantial stochastic variability.

To characterise the deterministic component of the signal, the irradiance was averaged across all years as a function of hour of day from Figure 2. The resulting "climatological" daily cycle is close to a smooth, unimodal curve: irradiance rises sharply after sunrise, peaks around local solar noon at approximately 350–400 W/m², and decays symmetrically towards sunset. This confirms a clear 24-hour seasonal component, well-suited to modelling with a SARIMA process with daily seasonality.

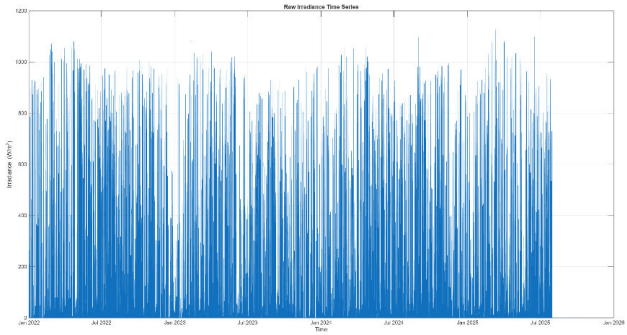

Figure 1: Raw Irradiance Time Series

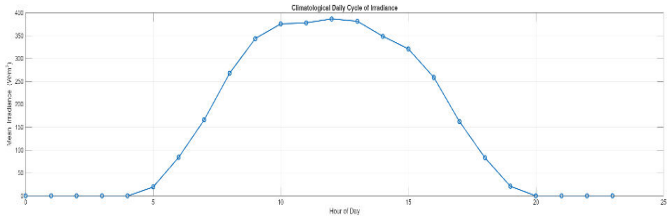

Figure 2: Climatological Daily Cycle of Irradiance average

The day-of-year vs. hour-of-day heatmap, Figure 3, clearly shows the interaction between the diurnal and annual cycles: summer days (DOY 170–230) have longer, brighter solar windows, while winter days have shorter, lower-irradiance periods. The smooth gradients in both directions indicate that compact seasonal encodings (e.g., sinusoidal hour and day-of-year terms) can capture most of this structure for SARIMA and QRF.

The daylight-only irradiance distribution is strongly right-skewed Figure 4, with many low-irradiance hours and a long tail towards clear-sky values above 900–1000 W/m². Together with the hour-of-day boxplots, Figure 5, which show largest spread around midday and low variability near sunrise and sunset. This reveals heteroscedastic, non-Gaussian behaviour. Consequently, probabilistic methods and quantile-based models such as QRF, evaluated with metrics like CRPS and prediction-interval coverage, are more appropriate than purely Gaussian point-forecast assumptions.

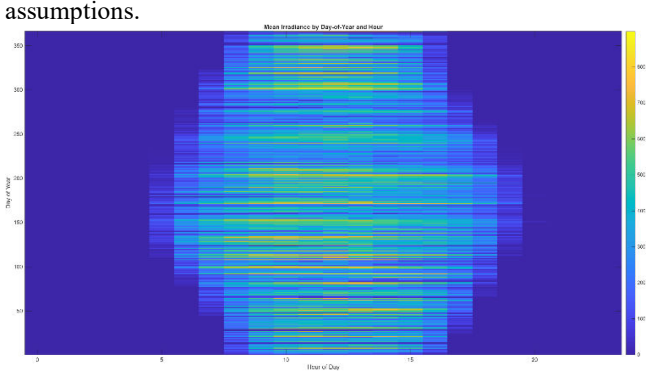

Figure 3: Mean irradiance by Day of Year and hour

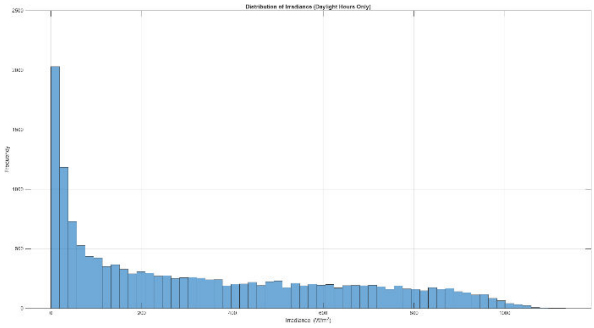

Figure 4: Distribution of Irradiance (Daylight Hours only)

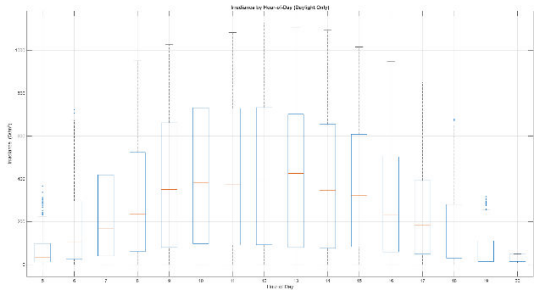

Figure 5: Irradiance by Hour of Day (Day Light only)

3.3. SARIMA MODEL DEVELOPMENT:

The autocorrelation of the hourly irradiance series, Figure 6, clearly shows a strong daily pattern. Correlations are high and positive for the first few hours, then oscillate between positive and negative values with peaks roughly every 24 hours and troughs around 12, 36, and 60 hours. This repeating wave tells us that irradiance at a given hour is related not only to the previous few hours, but also to the same hour on previous days. Because of this clear 24-hour cycle, a purely non-seasonal ARIMA model would struggle: it would need many lags to imitate the daily structure. Instead, it is more natural to use a Seasonal ARIMA (SARIMA) model, which is designed to handle exactly this kind of repeated pattern.

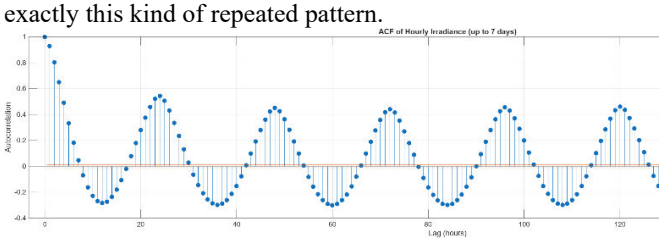

Figure 6: ACF of Hourly Irradiance (up to 7 days)

SARIMA extends ARIMA by adding seasonal terms that connect each value to the same time one day earlier. With a season length of 24 hours, the model learns both short-term effects (1–2 hours ago) and daily repetition. An Augmented Dickey–Fuller test on the cleaned irradiance data gave a p-value of 0.001, showing the series is already stationary, so no differencing is needed. Based on the autocorrelation pattern, we chose a simple SARIMA with one short autoregressive term and one seasonal term at lag 24.

The residual diagnostics confirm that this specification is adequate. The ACF of the SARIMA residuals Figure 7 no longer shows the strong 24-hour oscillations; most autocorrelations now lie within the 95 % confidence bounds and quickly decay to values close to zero. The partial ACF of the residuals from also shows no large spikes beyond the first lag. This means the main temporal structure in the data has been captured, and the remaining errors behave approximately like uncorrelated noise.

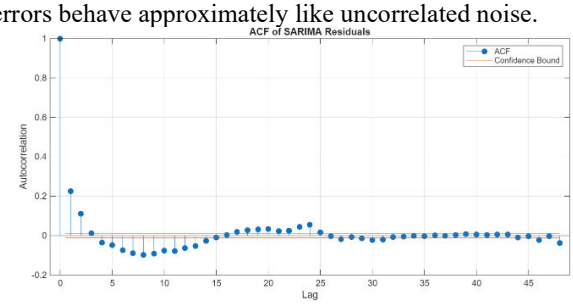

Figure 7: ACF of SARIMA residuals

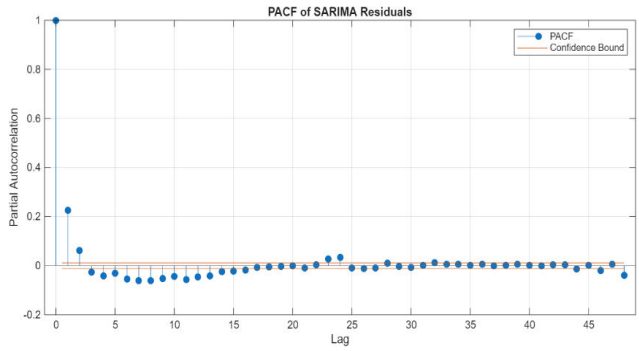

Figure 8: PACF of SARIMA Residuals

These results show that SARIMA provides a statistically sound and physically interpretable baseline model that captures the dominant 24-hour dependence in the irradiance series. By removing most of the temporal structure and leaving nearly white residuals, SARIMA offers a credible benchmark against which the added value of the QRF–based probabilistic forecasts can be evaluated.

3.4. Quantile Regression Forest (QRF) Specification

The Quantile Regression Forest (QRF) model serves as the ensemble machine learning counterpart. QRF is an extension of the Random Forest algorithm that is highly effective at non-linearly mapping a rich, robust feature set to the target variable. By leveraging exogenous meteorological information and noise-free temporal features, QRF is uniquely positioned to model the complex, stochastic fluctuations driven by instantaneous cloud cover and atmospheric conditions, thereby capturing the deviation from the deterministic solar trajectory.

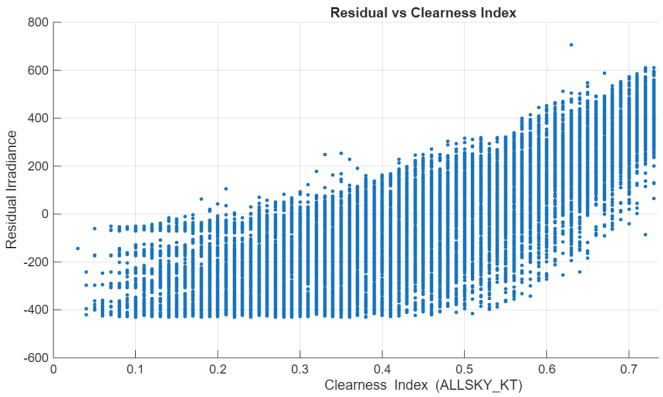

Figure 9: Residual Vs Clearness Index

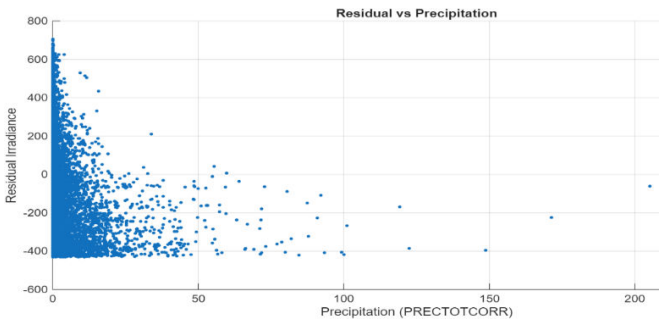

Figure 10: Residual vs Precipitation

The SARIMA residual vs clearness-index plot, Figure 9 shows a clear, nonlinear, and heteroscedastic pattern. For cloudy conditions (low KT), residuals are mostly negative with modest spread, while for clear-sky conditions (high KT), residuals become strongly positive and much more dispersed. This means the SARIMA model systematically over- or under-predicts depending on sky state, and its errors do not have constant variance.

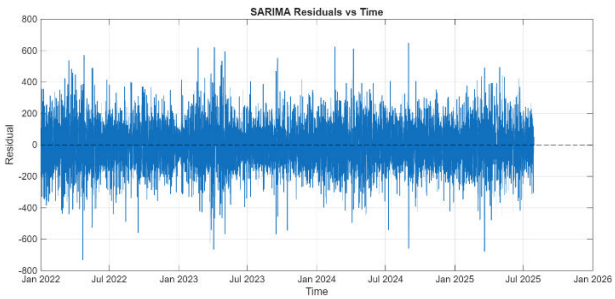

Figure 11: SARIMA Residual vs Time

The residual–precipitation plot, Figure 10 tells a similar story. Even small amounts of rain are associated with large negative residuals, indicating SARIMA over-predicts during precipitation events, and the relationship becomes highly irregular for heavier rainfall. These state-dependent, non-Gaussian error structures are complicated for linear parametric models but are well suited to Quantile Regression Forests, which can split on KT and precipitation and learn conditional, weather-dependent prediction intervals.

In Figure 11, the SARIMA residuals for the full training period are symmetric about zero and have no apparent long-term trend, which reinforces the message that this model has captured most of the systematic structure in this irradiance series. Aside from a few isolated spikes, the spread of residuals is relatively constant in time, implying roughly constant variance with some extreme forecast errors during periods of rapid change. The lack of any visible daily or seasonal banding in this plot reinforces the earlier ACF/PACF results by showing that the remaining errors are predominantly uncorrelated noise.

4. RESULTS AND DISCUSSIONS

We first estimated the SARIMA baseline using the full 2022–2025 training series, assuming a 24-hour seasonality. The Augmented Dickey–Fuller test returned a p-value of 0.001, confirming that the pre-processed irradiance series is stationary and may thus be modelled using linear time-series methods. The best-fitted SARIMA model reached an AIC of 364,432.1 and a BIC of 364,457.1, values that indicate a reasonable compromise between goodness of fit and model complexity for this large data set.

We compare the performance of SARIMA with an 8-feature QRF, using hour-of-day and day-of-year encodings, clear-sky irradiance, clearness index, and 24-hour lag/rolling irradiance as predictors for the day-ahead forecast. The metric comparison plot, Figure 15, shows that QRF outperforms the SARIMA baseline substantially on point-forecast accuracy. SARIMA yields an RMSE of 81.1 W/m² and an MAE of 47.8 W/m², while QRF reduces these errors almost by half to 41.3 W/m² and 21.3 W/m², respectively. The Continuous Ranked Probability Score (CRPS) improves from 54.2 for SARIMA to 15.0 for QRF, indicating much sharper and better-calibrated probabilistic forecasts.

Improvements of this nature are emphasized by the P50, or median, forecast trajectories shown in Figure 12. SARIMA captures the general shape of the bell-shaped diurnal profile but tends to underestimate the late-morning ramp and overestimate irradiance during the mid-afternoon cloud event. On the other hand, the QRF P50 curve traces both the smooth, clear-sky periods and the sharp dip around midday much more closely, reflecting the model's ability to exploit non-linear relationships between irradiance and relevant meteorological predictors, such as the clearness index and precipitation.

The prediction-interval plots show the probabilistic behaviour of both models. For SARIMA Figure 13, the 80 % interval defined by the P10 and P90 quantiles is relatively wide and almost monotonic in time, providing full 90 % coverage (Cov90 = 100 %) but at the cost of a

large, normalized interval width (PINAW = 0.669). This indicates a conservative but diffuse uncertainty description. The QRF intervals Figure 13 are noticeably narrower (PINAW = 0.154) yet still achieve high empirical coverage (Cov90 = 95.8 %). Furthermore, the QRF bands visibly widen during the cloudy mid-day period and contract during clearer hours, matching the heteroscedastic nature of solar variability.

QRF shows marginally better alignment at the lower probability level, and SARIMA at the upper level, but in general, both models yield useful probabilistic information. Results, summarized for 29 July 2025, show that including exogenous NASA meteorological features via QRF yields significant improvements in both point and probabilistic accuracy relative to a purely univariate SARIMA benchmark. SARIMA retains some value as an interpretable baseline that captures the dominant 24-hour persistence structure, but sharp, weather-driven fluctuations, which are most important for short-term PV forecasting and operational decision-making, are better captured by the data-driven QRF model.

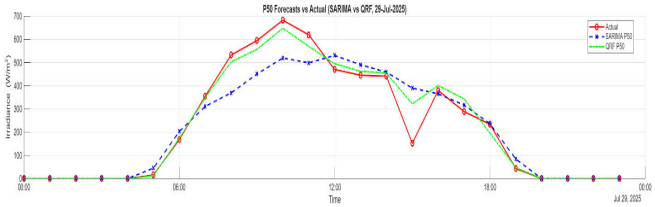

Figure 12: P50 Forecast vs Actual (SARIMA vs QRF 29 July 2025)

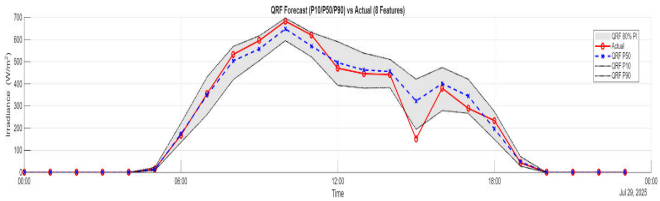

Figure 13: QRF Forecast (P10 P50 P90) vs Actual (29 July 2025)

Table 1: Performance Comparison

Model	RMSE	MAE	CRPS	Cov90 (%)	PINAW
SARIMA	81.1	47.8	54.1979	100.0	0.669
QRF (8F)	41.3	21.3	15.0089	95.8	0.154

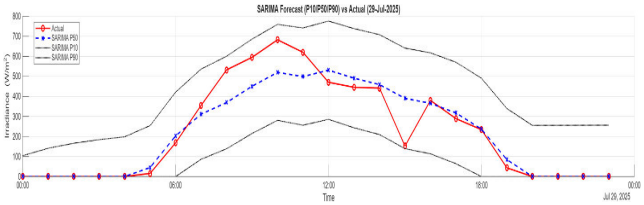

Figure 14: SARIMA Forecast (P10 P50 P90) vs Actual (29 July 2025)

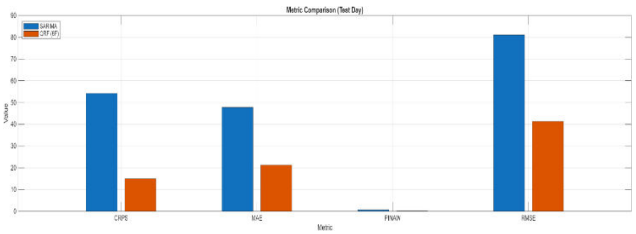

Figure 15: Metrics Comparision (Test Day)

5. CONCLUSION

Overall, the SARIMA model provides a statistically sound and physically reasonable baseline for day-ahead irradiance forecasting, against which the more flexible QRF model can be compared. The effect of the width of the prediction interval from an operational perspective is to determine how much reserve or flexibility the operator needs to schedule. Very wide 90% intervals (high PINAW) are produced by the SARIMA model, which, though conservative, implies that a large upward/downward reserve band must be kept available to cover uncertainty, effectively over-hedging, increasing operating cost. On the contrary, substantially narrower intervals were yielded by the QRF model, with close to the desired 90% coverage and better CRPS. It therefore means that, for the same reliability target, the operator can commit less reserve capacity, which translates into lower balancing costs and more efficient use of conventional generation or storage, while still being protected against most forecast errors.

REFERENCES

[1] M. Sun, C. Feng, and J. Zhang, “Probabilistic solar power forecasting based on weather scenario generation,” *Appl Energy*, vol. 266, May 2020, doi: 10.1016/j.apenergy.2020.114823.

[2] R. Basmadjian and A. Shaafieyoun, “Assessing ARIMA-Based Forecasts for the Percentage of Renewables in Germany: Insights and Lessons for the Future,” *Energies (Basel)*, vol. 16, no. 16, Aug. 2023, doi: 10.3390/en16166005.

[3] F. Eldali, T. Hansen, S. Suryanarayanan, and E. Chong, *Employing ARIMA models to improve wind power forecasts: A case study in ERCOT*. 2016. doi: 10.1109/NAPS.2016.7747861.

[4] S. Atique, V. Roy, and S. Bayne, *Forecasting of total daily solar energy generation using ARIMA: A case study*. 2019. doi: 10.1109/CCWC.2019.8666481.

[5] M. Sun and C. Feng, "Probabilistic solar power forecasting based on weather scenario generation," *Appl Energy*, vol. 266, p. 114823, May 2020, doi: 10.1016/j.apenergy.2020.114823.

[6] G. Tunnicliffe Wilson, "Time Series Analysis: Forecasting and Control,5th Edition, by George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel and Greta M. Ljung, 2015. Published by John Wiley and Sons Inc., Hoboken, New Jersey, pp. 712. ISBN: 978-1-118-67502-1," *J Time Ser Anal*, vol. 37, p. n/a-n/a, Mar. 2016, doi: 10.1111/jtsa.12194.

[7] P. Brockwell and R. Davis, *An Introduction to Time Series and Forecasting*, vol. 39. 2002. doi: 10.1007/978-1-4757-2526-1.

[8] D. Jurj, D. Micu, and M. Alexandru, *Overview of Electrical Energy Forecasting Methods and Models in Renewable Energy*. 2018. doi: 10.1109/ICEPE.2018.8559807.

[9] NASA, "NASA Open Data Portal." Accessed: Dec. 06, 2025. [Online]. Available: https://data.nasa.gov/dataset/prediction-of-worldwide-energy-resources-power